

HOMEWORK 6
NUMERICAL METHODS
FALL 2025

DUE FRIDAY, NOV 14

Instructions: Complete the following problem.

Computer Prob 1. A well known equation that is used to model phase-separation of materials is given by

$$u_t - \Delta u + \frac{1}{\epsilon^2} f(u) = 0, \quad \Omega = [0, 2\pi]^2.$$

where $f(u) = u^3 - u$. In 2D (two space dimensions), the solution will evolve into configurations that will generate transition layers connecting the values of -1 and 1 (representing pure material states in a binary mixture). Using the formal method of asymptotic expansions, it is possible to obtain the profile approximation of the so called diffusive interface for the above nonlinear PDE. The profile is found to be

$$u_o(z) = \tanh(z).$$

Task: approximate $u_o(z)$ on $[-3, 3]$ by using 11 equally space nodes and constructing $P_n(x)$. Then, you can repeat the process and use $S(x)$, cubic splines. Plot the original function, the classic interpolating polynomial and the cubic spline in the same figure. Describe your results. In addition, can you extend your work to $[-5, 5]$ - what happens in this case? Can you continue to increase the number of subintervals to decrease error?

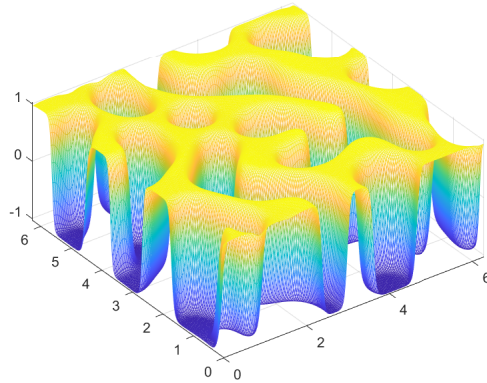


FIGURE 1. Numerical solution of the above model by using a random initial condition.

Note1: The above tasks are to be done with the codes developed in class: *NO built-in functions allowed or needed*. **Note2:** notation-wise $(n + 1)$ distinct points generate an interpolating polynomial of degree n , $P_n(x)$. $(n + 1)$ knots generate n sub-intervals each containing a polynomial of degree 3, and we refer to the spline as $S(x)$.

Important The programming should be done in terms of a Script and a function file in Matlab. Your submission should include a folder named “YourLastName” that contains 1) a .txt file with basic instructions/description of your script.m and your function.m file 2) script.m file and 3) function.m file. It is important to compress your folder before submitting it on Canvas. Try something like Winzip or Winrar. **In addition to the compressed folder with all content, you also need to submit a .pdf file which contains your homework problems in project format.**
